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Beyond $\alpha = 0.05$

A Data-Driven System for Alpha Level Selection
Master Thesis

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Hypothesis Testing

- Statistical hypothesis testing is widely used in empirical research.
- Researchers test whether an observed effect reflects a real relationship or random variation.

$$\text{p-value} \longrightarrow \text{Compare with } \alpha$$

$$p \leq \alpha \quad \Rightarrow \text{Reject } H_0$$
$$p > \alpha \quad \Rightarrow \text{Fail to Reject } H_0$$

[Fis25]

Why do we use $\alpha = 0.05$?

- α represents the probability of a Type I error.
- In practice, researchers often use $\alpha = 0.05$ as a standard threshold.
- This value became widely adopted as a historical convention.

[Fis25; CD82]

The Problem with $\alpha = 0.05$

One Size Fits All

Same threshold applied regardless of

- sample size
- effect size

Cliff-Edge Effect

- $p = 0.049 \rightarrow$ Significant
- $p = 0.051 \rightarrow$ Not significant

Replication Crisis

- Many statistically significant findings fail to replicate.

[Ope15; WL16]

Existing Approaches for Choosing α

- Fixed significance levels (0.10, 0.05, 0.01).
- Power-based approaches considering sample size and effect size.
- Decision-theoretic approaches based on error costs.
- Recent proposals suggest stricter thresholds such as $\alpha = 0.005$.

[Coh88; Ber85; Ben+18]

Limitations of Existing Approaches

- Most approaches still rely on fixed thresholds.
- They do not adapt the significance level to study characteristics.
- There is no widely used practical system for selecting α.

Research Objective

- Build a system that recommends a Study specific (α).
- The recommendation is based on study or research context.
- Key factors include sample size, effect size, risk tolerance, test type, and number of hypothesis tests.
- These factors influence statistical power and error probabilities in hypothesis testing.

[Coh88; Bak16; HFC19]

Research Question

Can a data-driven and context-sensitive approach improve the choice of the significance level compared to the conventional value $\alpha = 0.05$?

Application Overview

- The system recommends a study-specific significance level.
- The recommendation is generated in two stages.
  - a data-driven baseline α prediction.
  - context-based adjustments.

System Architecture

INPUT		STAGE 1		STAGE 2		OUTPUT
Sample Size + Effect Size Sample size Effect size (known pre-study)	→	Random Forest Regression Trained on 664 replication studies MAE-optimized	→	Context Adjustment Risk tolerance Test direction Multiple testing	→	Final α Recom-mendation Clamped to 0.001 – 0.15 Ready to use

$$\alpha_{final} = \alpha_{baseline} \times k_1(\text{risk}) \times k_2(\text{direction}) \times k_3(\text{multiple testing})$$

Replication Dataset

- Data collected from replication projects available on OSF.
- studies across fields such as **biomedical research**, **psychology**, and **social sciences**.
- Includes **p-values**, **sample sizes**, and **effect sizes**.
- Final dataset contains **664** hypothesis tests.

[Cha25; ED21; Sot17; Sot22; Li21; Ope15]

Example Dataset Rows

Sample Size	Effect Size	p-value	Optimal α
120	0.35	0.041	0.052
85	0.28	0.063	0.061
241	0.30	0.048	0.074
150	0.22	0.071	0.066
96	0.40	0.032	0.049

- Each row represents one hypothesis test.
- Sample size and effect size are used as input features.
- Optimal α is the target variable used for model training.

Optimal Alpha Definition

Replicated

Keep original α
(decision validated)

Failed

Adjust α: largest
$\alpha < p$-value
(correct decision)

Example

$p = 0.047 \rightarrow$ failed to replicate
Optimal $\alpha = 0.04$ (not 0.05)

Features Used in the Model

- Sample size
- Effect size
- Optimal Alpha (Target Variable)

Predicting Baseline α

Machine Learning Task

- Goal: predict the baseline significance level (α).
- Each hypothesis test is treated as one observation.
- Input features:
  - Sample size
  - Effect size
- Target variable: optimal α derived from replication outcomes.

Algorithms Tested

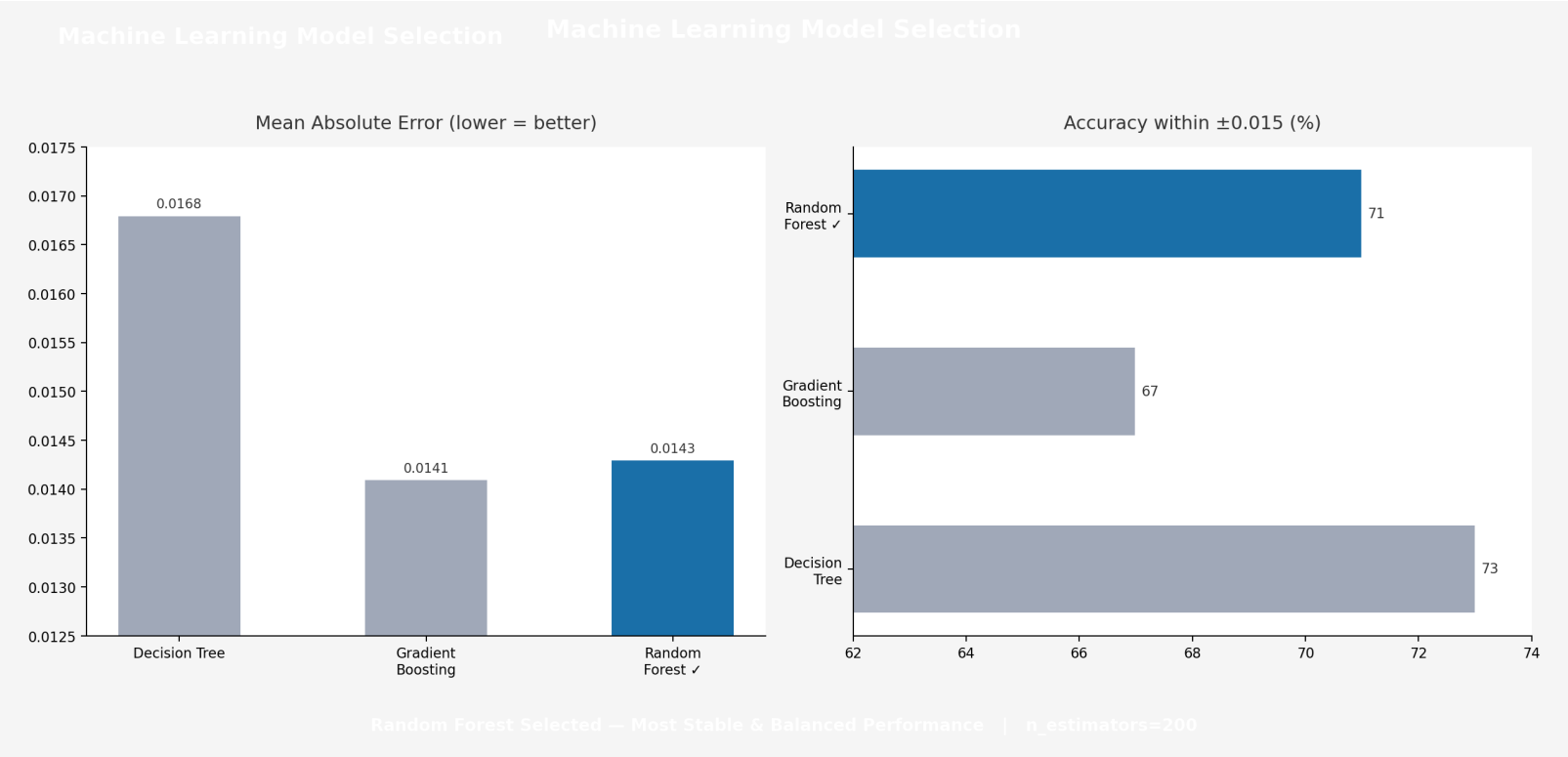
Regression Models Evaluated

- Decision Tree Regressor
- Gradient Boosting Regressor
- Random Forest Regressor

Evaluation Metric

Mean Absolute Error (MAE)

Machine Learning Model Selection



Performance comparison of Decision Tree, Gradient Boosting, and Random Forest models.

Selected Model: Random Forest

Final Model

- **Random Forest Regressor** selected as the final model.
- Achieves strong performance in both **error reduction** and **prediction stability**.
- Handles non-linear relationships between study characteristics and optimal α.

Implementation

Random Forest with **200 trees** used for baseline α prediction.

Context-Based Adjustment

Motivation

- The machine learning model predicts a **baseline significance level**.
- However, the final decision threshold should also reflect the **research context**.

Adjustment Factors

- Risk tolerance
- Test direction (one-tailed vs two-tailed)
- Number of hypothesis tests (multiple testing)

Stage 2: Context-Based Adjustment

k_1 — Risk Tolerance

- Very Strict $\times 0.50$
- Strict $\times 0.75$
- Neutral $\times 1.00$
- Lenient $\times 1.25$
- Very Lenient $\times 1.50$

k_2 — Test Direction

- Two-tailed $\times 1.00$
- One-tailed $\times 2.00$

Reflects rejection region allocation.

k_3 — Multiple Testing

Bonferroni correction:

$$\alpha/m$$

where m = number of hypothesis tests.

Final clamping: $0.001 \leq \alpha_{final} \leq 0.15$

Final Alpha Calculation

Adjusted Significance Level

$$\alpha_{final} = \alpha_{baseline} \times k_1(\text{risk}) \times k_2(\text{direction}) \times k_3(\text{multiple testing})$$

Interpretation

- $\alpha_{baseline}$: predicted by the machine learning model.
- k_1: adjustment for risk tolerance.
- k_2: adjustment for test direction.
- k_3: adjustment for multiple hypothesis testing.

System Deployment

Implementation

- The alpha recommendation system is implemented in **Python**.
- Machine learning model developed using **scikit-learn**.
- An interactive interface built using **Streamlit**.

System Functionality

- Users provide study characteristics as inputs.
- The system predicts a baseline α using the trained model.
- Context-based adjustments are applied to produce the final recommendation.

User Inputs

Required Inputs

- Sample size
- Expected effect size
- Risk tolerance level
- Test type (one-tailed or two-tailed)
- Number of hypothesis tests

Purpose

These inputs describe the study characteristics and research context used to compute the recommended significance level.

Alpha Recommendation System Interface

Example interface of the Streamlit-based system used to generate a recommended significance level.

Evaluation Setup

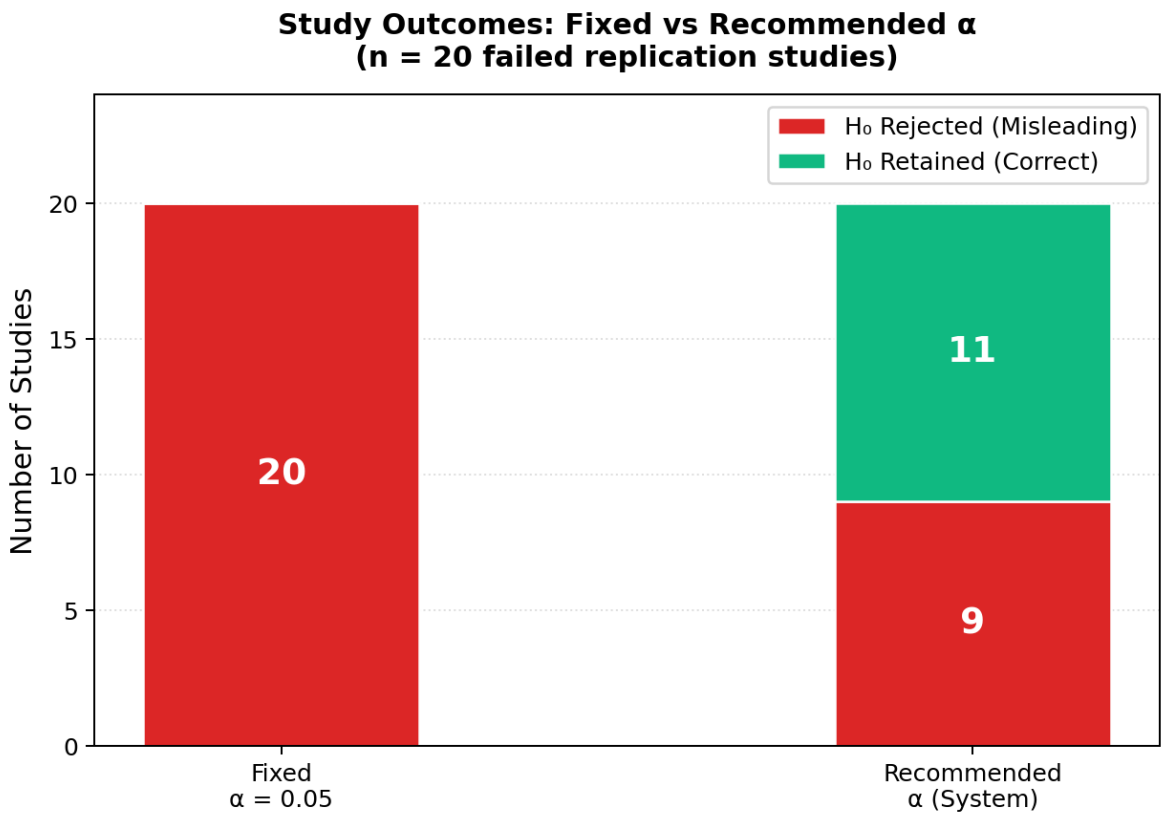
Testing Procedure

- Evaluation conducted on **20 failed replication studies**.
- Two decision rules were compared:
  - Conventional threshold: $\alpha = 0.05$
  - Recommended threshold: α generated by the system
- For each study, the hypothesis decision was examined under both thresholds.

Objective

Assess whether the recommended α reduces misleading rejections in failed replication studies.

Main Findings



Comparison of study outcomes using the fixed significance level ($\alpha = 0.05$) and the recommended significance level produced by the system.

Limitations

Study Limitations

- Dataset derived from a limited number of replication studies.
- Study characteristics used as inputs are restricted to a few key variables.
- Model predictions depend on patterns observed in the available data.

Implication

The recommended significance levels should be interpreted as guidance rather than a universal replacement for fixed thresholds.

Conclusion

Key Takeaways

- The conventional significance level $\alpha = 0.05$ may not be appropriate for all studies.
- A data-driven approach can provide context-sensitive recommendations.
- The proposed system combines machine learning with contextual adjustments to recommend a suitable α.

Future Work

Possible Extensions

- Expand the dataset with additional replication studies.
- Incorporate more study characteristics as input features.
- Evaluate alternative machine learning models.
- Deploy the system as a publicly accessible research tool.

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Vielen Dank

für Ihre Aufmerksamkeit

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